

Recommended PET Configuration

Comparison with Gregor's Implementation

INTERIM INVENTORY — the selection is not yet made

Publication Agent B (PET)

18 September 2026

This deck inventories differences. It does not select a configuration.
The matched comparison that would select one is specified on slides 14–16 and in
MATCHED_COMPARISON_PROPOSAL-20260918.md. Nothing is launched.
PET is diagnostic method development, not a publication product.

Where the selection stands

No configuration is selected, and this deck cannot select one. It compares components; the decision needs one matched comparison of the two *complete* configurations on our unfolding task.

what we have	what it supports	what it does not
a pinned, code-traced inventory of every consequential difference	knowing <i>what</i> differs and which differences are cheap to change	any claim that either configuration unfolds better
component measurements on a synthetic fixture and on GPU cost	cost, and one inconclusive routing result	transfer to the real task — direction and magnitude do not carry
a production baseline measured on the real endpoint (recovery 0.5126)	a scale, a margin, and a frozen protocol to reuse	a value for <i>his</i> configuration — it has never been run on our task

Provisional engineering choice, stated as such: retain ours — it is frozen, validated, cheaper, and its closure receipts exist. Retention on continuity and cost is not a demonstrated result, and this deck does not upgrade it into one.

What would settle it: slides 14–16.

What is pinned, and what “ours” means

label	what it is	pin
ours, production	pet-fullevent-fps-v1: the estimator the driver stamps	44142e8c (branch pet-direct-token-comparison)
ours, experimental	four-arm representation study: typed descriptors, routing, aggregate overflow. No arm is in the production path.	44142e8c, direct_token_comparison/
ours, proposed	what this deck recommends changing	nothing is built
Gregor's	gregorkrz/minerva-ml	fc9a099d, 2026-07-20, clean tree

- ▶ **fc9a099 is the repo as cloned, not a paper commit.** No release tags; the repo has evolved past arXiv:2604.12364. The paper commit must come from Gregor.
- ▶ Everything attributed to him is the **default** unless marked optional. Verified at fc9a099: use_pid on, use_cond on, include-E-sum on, max_particles 33, **use-max-blobs-and-prongs off** — so his overflow aggregation is optional and not the default.
- ▶ **He is not doing unfolding.** Supervised regression/classification, MC only, no data leg, no per-event weights. Several rows below have no counterpart, and that is a difference in problem, not in quality.

The actual baseline: our production estimator

Data path

- ▶ cloud = **raw non-muon calorimeter clusters** ($E, \text{pos}, z, \text{view}, t$), 5 columns
- ▶ muon **removed from the cloud**; carried as a 13-feature event block, FiLM conditioning
- ▶ cap: stable energy sort, **top 12**, zero-pad — applied **at dump time**, so changing it needs a re-dump
- ▶ overflow **discarded with no trace**: no count, no summed energy, no flag
- ▶ fixed physical divisors; event features z-normalized with *frozen* reco-MC pass_reco statistics
- ▶ azimuth as $(\cos \phi, \sin \phi)$ everywhere

Model and training

- ▶ PET: encoder $\rightarrow$ **KNN local aggregation** ($K = 3$ over (pos, z)) $\rightarrow$ 2 blocks, 2 heads, width 32, LayerScale $\rightarrow$ class token + FiLM head
- ▶ **47,041 trainable parameters**; 93,954 per OmniFold iteration
- ▶ weighted BCE; Adam 10^{-4} annealed to 10^{-5} , realized LR *asserted* at runtime; batch 512; niter 3, epochs 8; 2 M subsample
- ▶ MultiFold; push weight $w = \exp(\text{logit})$ in logit space
- ▶ background: negweight-refined, fail-closed
- ▶ seed 42 frozen — this is what makes C_{stat} statistical

`train_fullevent_nominal.py:403-407, 47-71; fullevent_fps_dataloader.py; dump_pointcloud_inputs.py:90-112; omnifold_nn/omnifold/{net,omnifold}.py; receipts/model-capacity.json`

Data path

- ▶ cloud = **reconstructed objects**: muon, ≤ 2 photons, blobs, prongs — 10 columns each
- ▶ η , ϕ , $\log p_T$, $\log E$, **PID code** (8-class learned embedding), dE/dx , x, y, z, t (all $/10^4$)
- ▶ muon is **a token like any other** (PID 0)
- ▶ caps in series: 150 at preprocessing (energy-sorted *only* if exceeded), then **first 33 rows at batch time, no re-sort**
- ▶ absent fields **zero-filled, no validity mask**
- ▶ 16 global features: recoil calorimetry, passive material, Michel, $\gamma\gamma$ mass, prong count, **6 per-PID log energy sums**

Model and training

- ▶ PointGlobalMixedViT: mixed encoder $\rightarrow$ **Fourier positional MLP over** $(\eta, \phi) \rightarrow 4$ pre-norm blocks, 8 heads, $d_{\text{model}} = 128 \rightarrow$ CLS fused with EVT token
- ▶ **890,130 parameters** — $18.9\times$ **ours**
- ▶ Huber on $\log 1p$; **AdamW**, wd 0.01, **1000-step warmup then cosine, grad clip 1.0**; batch 2048; 100k–500k steps
- ▶ optional: OmniLearned PET2, HyperScale, BERT, pretrained init
- ▶ **pretrained checkpoints are unreachable** — HyperScale repo 404, NERSC URLs time out (audit B_provenance)

vit.py:226-373; train.py:465-756, 2328-2358; preprocessing.py:257-583; jobs/submit_train_jobs.py:150-151;
gregor-audit/jobs/A_dataset.out.md

Comparison I — representation

	ours	Gregor's	consequence	recommendation
objects	raw non-muon calorimeter clusters; muon distinguished	reconstructed objects; muon is a token	mean 74.6/102.6 clusters vs 14.0/15.6 objects: $5.3\times/6.6\times$ fewer tokens. Whether they carry the same information is unmeasured	borrow the vocabulary; keep the distinguished muon
token features	5 cols, no type code; mask = ($E=0$)	10 cols incl. direction, dE/dx , time, PID embedding	more physics per object; but zero-fill hides absence, and padding shares PID index 0 with muon (masked, so latent)	borrow features + embedding; reserve index 0 for padding
transforms	physical divisors; frozen MC z-norm; $(\cos \phi, \sin \phi)$	logs $+/10^4$; LayerNorm inside encoder; raw ϕ	raw ϕ is discontinuous at the wrap; we fixed this in July	keep ours; recommend the ϕ fix to him
event globals	13, all muon/vertex	16, recoil calorimetry + 6 per-PID energy sums	his sums are a fixed-width channel that survives a cap — but he computes them <i>after</i> truncation	borrow the sums, computed <i>before</i> any cap
caps and overflow	top 12 by energy, rest discarded silently	150 then first 33 unsorted; aggregate token <i>optional</i> , off	our cap binds 64%/84% vs his 9–17%/11–27%; his ordering is worse	borrow aggregation; keep energy ranking

Comparison II — model, training, integration

	ours	Gregor's	consequence	recommendation
routing	one family, so no question in production; pooled vs individual measured experimentally	individual, unconditionally	no accuracy difference detected; individual costs $1.24\times$ at the operating point, $1.88\times$ in the tail	keep pooling as a practical default
architecture	2 blocks, 2 heads, width 32, KNN in (pos, z) ; 47k params	4 blocks, 8 heads, width 128, Fourier PE in (η, ϕ) ; 890k params	$18.9\times$ gap; ours is retrained ~ 130 times per campaign, his once	keep architecture; treat capacity as open
init	random, seed 42 frozen (this makes C_{stat} statistical)	random; optional pretrained OmniLearned	checkpoints unreachable: 404 and timeouts	keep; ask Gregor for the weights and hashes
training	Adam $10^{-4} \rightarrow 10^{-5}$ step anneal; no wd, no clip	AdamW, wd 0.01, warmup + cosine, clip 1.0	judgement, unmeasured on our estimator: his schedule is better conditioned, and free. wd biases a likelihood-ratio classifier ($w = e^{\text{logit}}$)	borrow warmup/cosine/clip; not weight decay
OmniFold	full MultiFold, logit-space ratio, negweight background, fail-closed data path; closures PASS	no counterpart — MC only, no weights, no flags	his dataset cannot be an OmniFold input; his vocabulary can	keep ours in full

The constraint that decides what can be compared

Every distinctive element of his representation is absent from our production input — for every inventory.

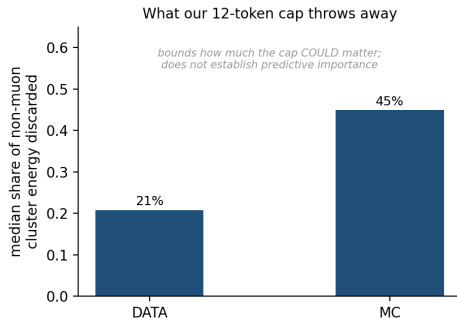
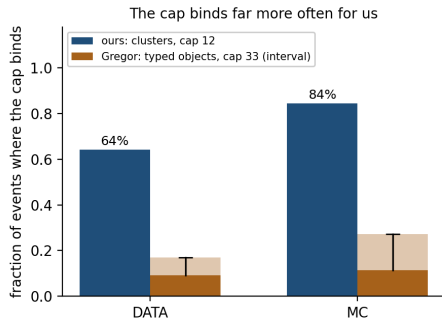
The G2 dump writes `part_reco`, `part_gen`, `reco_scalars`, `truth_scalars`, `reco_muon`, `reco_vertex`, `reco_view`, `reco_time`, `pass_reco/pass_truth`, weights and provenance. There is **no** blob, prong, photon, PID, dE/dx , Michel, per-type-energy-sum or overflow-telemetry key anywhere in it. Re-measured at 44142e8c; first established by the audited Gregor-PET2 campaign in July (absent/absent/absent/absent).

tier	what it compares	prerequisite	status
A	architecture + training recipe, both arms on the frozen G2 input	none	runnable
B	representation: typed objects, PID embedding, 16 globals, cap 33, aggregate overflow	a symmetric data/signal/background typed-object dump with overflow telemetry	blocked

Tier B's prerequisite is a C++ event-loop deliverable, requested in `FULL_EVENT_INTERFACE_REQUEST.md §D`, not delivered, owned by the active-universe C++ owner — **not schedulable by me and not costed**. His public rows are MC-only, so they cannot supply a data leg; a truth-derived type vocabulary is excluded permanently as leakage.

Tier A is the matched comparison we can have, and it is worth having. **A Tier-A result may never be written**

The measured problem: our cap binds, and it binds hard



Unselected, unweighted tuple entries: 17,930 data (1B) + 186,439 MC (1A). Not the selected population.

Same two tuple files, both vocabularies. Our 12-token cluster cap binds in **64.1 % (data) / 84.4 % (MC)** of events and discards a median **20.8 % / 44.9 %** of non-muon cluster energy. A 33-token typed-object cap binds in **9.2–16.9 % / 11.4–27.2 %** — an interval, because the receipt holds marginal, not joint, family multiplicities.

Three things this does not show. It does not show the discarded clusters carry predictive information. It does not predict a closure improvement — **no closure measurement of aggregation exists**. And these are **unselected, unweighted** tuple entries, so they do not determine selected-population behaviour.

`receipts/typed-object-budget.json; source-multiplicity.json (job 58470099)`

Measured accuracy and stability: routing is inconclusive

Eight paired seeds, four-object synthetic fixture,
information content / model size / budget / init / seeds
held identical between arms.

paired improvement, individual over pooled	
median	+9.7 %
mean	+0.8 %
seed-to-seed sd	25.8 points
95 % interval on the mean	$[-20.7, +22.4]$ %
paired p	0.50
favourable seeds	6 of 8

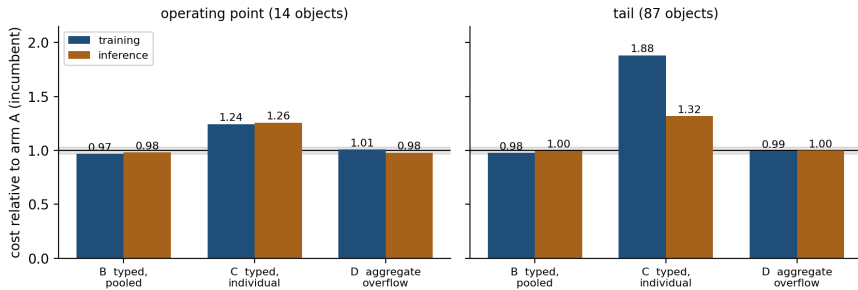
Frozen decision: NO_PASS, 142 of 146 checks holding.
Two failures *were* the inconclusive result; separately
shuffle-71 missed two projection checks marginally
(0.0119 vs 0.01; absolute errors well inside 0.05).

What each number measures —

- ▶ the interval is on the **mean paired injected-improvement percentage across training seeds on a synthetic fixture**. Not on production closure, not on real data.
- ▶ $p = 0.50$ is a paired test **across training seeds**. It measures seed scatter. **Test-sample uncertainty was not separately estimated**.
- ▶ 25.8 points is a **planning assumption** carried from a different fixture, two arms and a different denominator — **not measured power for the endpoint now specified**.

And: in the fixture every generic slot is noise while in production the cluster cloud carries most of the information. Neither direction nor magnitude of a synthetic effect transfers automatically.

Measured cost: what each change actually buys and spends



Grey band: $\pm 3.3\%$, the resolution of this measurement, estimated from the arm-A/arm-B null contrast. A bar inside it is not a difference.

One A100, TF32 off, determinism on, float32. Cost is cost: a cheaper arm is not a better one.

Aggregate overflow is *not* free — it is indistinguishable from parity at this measurement's resolution. Arm B is computationally identical to arm A in the pooled configuration and still reads $0.97\times$, so the floor is about $\pm 3.3\%$. The non-compute cost is real: a dump regeneration, a schema field, a re-validation, and a new estimator fingerprint.

Individual routing is outside that band and grows with multiplicity ($1.24\times \rightarrow 1.88\times$ training). Pooling is nearly flat. Our campaign multiplier is ~ 130 trains, so per-train cost is multiplied, not absorbed.

`cost-half.json` (one A100, TF32 off, determinism on, float32); `inference-benchmark.json`

Component inventory — all provisional, none selected

borrow (candidate)	why	cost	prerequisite
per-family log energy sums, over the whole event	cap-proof fixed-width channel	est. ≈ 0	fingerprint change; absent from the dump
aggregate overflow, energy-weighted merge	addresses a measured 21–45 % energy loss in most events	<i>measured</i> parity to $\pm 3.3\%$	dump regeneration; no closure evidence yet
richer per-token features + type embedding, index 0 reserved	more physics per token, fewer tokens	unmeasured	absent from the dump
warmup + cosine, gradient clipping	better-conditioned schedule	est. ≈ 0	Gate-4 fingerprint; closure re-run
one capacity step	18.9 $\times$ gap is a live hypothesis	multiplies through the UQ campaign	price against the campaign, not one train

These rows are component judgements, not a selection, and the single-factor experiments they imply are **RETIRED** — the four live items interact, so they are compared as complete configurations instead (slides 14–16). The rows survive as the record of what differs and as the source of the tuning grids.

Back to Gregor: encode ϕ periodically; reserve embedding index 0 for padding; map raw prong PID 9 (it currently raises); compute the per-PID energy sums before truncation.

Limitations — what these numbers do not say

1. “Negligible measured overhead” is not “free”. Aggregate overflow is inside the $\pm 3.3\%$ resolution of the cost measurement, and it carries a dump regeneration, a schema change, a re-validation and a new fingerprint.
2. Discarded energy does not establish closure improvement. 21–45 % bounds how much the cap *could* matter. No closure measurement of aggregation exists on either side.
3. Validation is incomplete: 10 of 13 widths released. Three widths failed the finite-difference check V5 and were not released. Diagnosis: the *criterion* was wrong, not the gradient — error falls as h^2 (1.8×10^{-2} at the frozen $h=10^{-2}$, 3.1×10^{-4} at 10^{-3}) then rises, the signature of truncation giving way to cancellation. **The criterion was not relaxed after the failures.**
4. Unselected source measurements do not determine selected behaviour. Two files, unweighted, no production selection, no POT weighting.
5. Every statistical quantity is labelled by what it measures — seed scatter, not test-sample uncertainty; a synthetic fixture, not production; a planning assumption, not measured power.
6. The pin is the repo, not the paper. fc9a099 may not be what produced arXiv:2604.12364.

One engine, and an endpoint that a null estimator fails

One engine. Both arms run inside our TensorFlow MultiFold. The prior campaign declined a cross-framework comparison because it would “confound framework, representation, and training engine”. One engine removes two of those confounds; Tier A removes the third by holding the representation fixed.

Gate V-PORT. His architecture is ported to Keras-2, **his PyTorch weights are transferred in**, and outputs must agree to $\leq 1e-5$ max absolute difference in float64 over 1,024 fixed inputs including fully-masked and single-token rows, with **equal** parameter counts. **A port that cannot be shown to compute his function is not his configuration**, and nothing downstream repairs that. Two mappings are forced and travel with every result: his Fourier positional encoding is fed (pos, z) because our tokens carry no per-token (η, ϕ) ; and his log $1p$ Huber is an E_{avail} regression loss with no counterpart in a reweighting classifier, so **the objective stays ours**. His arm keeps his optimizer, schedule, clipping and batch.

One endpoint — and deliberately not the cheap one. The ordinary self-consistency closure is **unusable**: it is an identity check that a *constant* estimator optimizes — structural zero power. Choosing it would let measurability pick the specification. The primary endpoint is the **powered injected truth-reweight recovery** $1 - E_w[|1 - r_b|]$ over 285 cells, on the protocol predeclared 2026-08-05 and reused *unchanged*: tilt amplitude 0.35, clipped $|z| \leq 3$, rate-preserving, truth-passing rows only, two **disjoint** 2 M halves at split seed 7 — frozen six weeks ago by someone who could not have known this comparison would use it. Our baseline: **0.5126033**; **his has never been run on it.**

The margin, and the rule that cannot be softened

Non-inferiority margin $\delta = 0.018$ recovery points, fixed before any number exists.

0.0180209 is *exactly* the margin by which our estimator clears its adopted criterion (recovery **0.5126033** vs **0.494582** = $0.80 \times$ ceiling 0.618228), so a larger deficit costs the entire margin by which production passes its own gate — decision-relevant, not a round number. It is $14.7 \times$ the closure's measured repeat spread ($1.226e-3$) yet only **2.9 %** of the ceiling: strict, not lenient.

95 % CI for mean(d), $d = \text{ours} - \text{his}$	conclusion
lower bound $> -\delta$, all safeguards pass	Recommend OURS — non-inferior by a margin that matters, at equal tuning, on our task
upper bound $< -\delta$, his safeguards pass	Recommend HIS — materially better; adopt it for method development
CI contains $-\delta$	Inconclusive — no selection. Incumbent retained as a <i>provisional engineering choice</i> , in those words
any safeguard fails for an arm	that arm is not recommendable at any endpoint value

A CI that merely contains zero decides nothing — only a lower bound strictly above $-\delta$ licenses “ours”. **Ours does not win by default.** The result is reported whether or not it favours us.

Per BEN-038, both arms also report signed response and scatter penalty: **an arm can win this aggregate purely by being less noisy per cell**, and that will be named as such.

Stages, costs, and what stops it

stage	GPU-h	gate to proceed
1 port to Keras-2; V-PORT gate; tests	0.5	$\leq 1e-5$, equal parameter counts
2 cost calibration, one evaluation per arm	3	measured cost replaces the estimate
3 tuning: 4 trials each , own default as trial 1	12	both arms frozen; halves untouched
4 variance pilot, 4 paired seeds $\times$ 2 arms	12	measured σ_d admits δ
5 final comparison, 8 paired seeds $\times$ 2 arms	24	—
contingency (one stage repeated)	18	
ceiling requested	≤ 70	of 290; 16.0 consumed

Sizing. Paired on seed. Prior $\sigma \approx 0.008$ (48-seed closure-adjacent spread) $\Rightarrow \sigma_d \approx 0.011$, $n \approx 3$; 8 seeds tolerates σ_d to ≈ 0.019 . **An assumption from an adjacent statistic, not measured power — stage 4 replaces it.**

Stops: V-PORT fails; the pilot cannot resolve δ inside the ceiling (**say so — do not raise the ceiling or widen δ**); a safeguard fails; limits exhausted; or the endpoint, margin, rule or samples would have to change. Repaired submissions count as retries; no automatic relaunch.

Frozen before stage 5, digest-verified: both configurations, the injection protocol, the sample, δ , α , the rule verbatim, every threshold. **Not relaxed after failures.**

What the final deck will lead with

Once stage 5 returns, the deck is restructured so its first three slides are:

1. **the selected configuration**, named in one line, with the word “selected” reserved for a case the decision rule actually licenses — and the words “provisional engineering choice” used instead when it does not;
2. **the matched evidence**: the paired recovery difference and its 95 % interval against δ , the signed-response / scatter split that attributes the difference, and the safeguard table;
3. **the conditions under which the conclusion holds** — Tier A only, so architecture and training recipe and *not* representation; one endpoint; $\delta = 0.018$; `fc9a099d` not established as the paper commit; pretrained initialization untested because unreachable; recovery measured against a reference ceiling that is not a proven bound.

The present inventory becomes the appendix. Two sentences are pre-committed now, so neither can be softened later:

If his configuration wins by more than δ , the deck recommends his and says so in the title.

If the interval is wide, the deck says the comparison failed to decide and names the incumbent as retained on cost and continuity — not as validated.

Open regardless of the outcome: the representation (Tier B, blocked on the dump), his published configuration (R1), and pretrained initialization (R2). **Questions for Gregor, not compute**: R1 paper commit, R2 checkpoints and hashes, R3 his E_{avail} definition. **Nothing has been sent to anyone.**

Appendix A — claim-to-evidence index

claim	evidence
cap binds 64.1 % / 84.4 %; median 20.8 % / 44.9 % energy discarded	direct_token_comparison/local_validation/20260918-source/source-multiplicity.json (job 58470099; 17,930 data + 186,439 MC entries)
typed cap binds 9.2–16.9 % / 11.4–27.2 %; 5.3×/6.6× fewer tokens	receipts/typed-object-budget.json (derived; 8 tests)
47,041 vs 890,130 parameters	receipts/model-capacity.json (both models instantiated)
routing: median +9.7 %, [−20.7, +22.4] %, $p = 0.50$, NO_PASS 142/146	local_validation/20260917-matrix/matrix-summary.json
inference 1.283×, all 5 checks pass, $CV \leq 2\%$	local_validation/20260917-inference/inference-benchmark.json
GPU cost by arm; A/D parity; $\pm 3.3\%$ floor	local_validation/20260918-a3-cost/cost-half.json
10 of 13 widths released, no hard stop	local_validation/20260918-tailval/validation-final.json; criteria sha256 abccd88b...
Gregor token format, PID scheme, caps, globals	gregor-audit/jobs/A_dataset.out.md (line-cited to fc9a099)
checkpoints unreachable; OmniLearned MIT;	gregor-audit/jobs/B_provenance.out.md
dataset CC-BY-4.0	
our production configuration	FULL_EVENT_FEATURE_CONTRACT.md; train_fullevent_nominal.py:47–71, 403–407

Full index: ../CLAIM_EVIDENCE_INDEX-20260918.md. Full inventory: ../CONFIGURATION_COMPARISON-20260918.md.

Appendix A2 — anchors for the matched comparison

claim	evidence
baseline recovery 0.5126033 against the adopted criterion 0.494582 = $0.80 \times$ ceiling 0.618228, margin +0.0180209	job 56552326, independently finalized by 56562169; docs/orchestration/CLAIM-CLM-012.md (viii)
closure repeat spread 1.226e-3 — same-configuration, <i>not seed scatter, so it sizes nothing</i>	job 56611837; CLAIM-CLM-012.md (ix)
97.8 % of the baseline's gap to the ceiling is scatter; signed bias -0.0019	FINDINGS-ARCHIVE-2026-08.md BEN-038
the ceiling 0.618228 is a <i>reference curve, not a proven bound</i> (a band was measured at $E_w[r] = 1.0333$)	CLAIM-CLM-012.md caveat (i); BEN-038
the ordinary closure has <i>structural zero power</i> injection protocol predeclared 2026-08-05, reused unchanged	AUDIT-FINDINGS-20260728.md; closure_fullevent_fps.py:4-9 closure_powered_truth_reweight.py:10-33
evaluation sample: 49.15 M signal / 4.12 M data / 0.56 M background rows, sha256 fa6b3463..., 9.90 GB	g2_fullevent/input/G2_FPS_MEFHC_P12_RECEIPT.json, status PASS, job 56120687
<i>his representation is absent from every inventory</i>	dump_pointcloud_inputs.py:190-235 at 44142e8c (re-measured); GREGOR_PET2_OMNIFOLD_ASSESSMENT.md:249-255 at b65f9ff2
seed-scatter prior $\sigma \approx 0.008$	CLAIM-CLM-010.md: 48 seeds, mean 0.014256, sd 0.008023 — <i>an adjacent statistic, a planning assumption only</i>
a PyTorch OmniFold backend exists (10.9 k lines) but its model is an <i>independent</i> reimplementation, not his	nd-unfolding/pet2_torch/ at b65f9ff2; model.py docstring

Appendix B — the V5 diagnosis, in full

Three widths — (0, 64, 2) and (2, 160, 12) for the individual arm, (2, 12, 4) for both — failed the finite-difference check, and only that check. My first hypothesis (a small denominator) was **wrong**: the failing coordinates have $|\nabla| \approx 0.0215$. Sweeping the step at the worst coordinate:

	h	numeric derivative	relative error
	3×10^{-2}	-0.01787	1.69×10^{-1}
1×10^{-2} (<i>frozen</i>)		-0.02111	1.84×10^{-2}
	3×10^{-3}	-0.02147	1.67×10^{-3}
	1×10^{-3}	-0.021497	3.14×10^{-4}
	3×10^{-4}	-0.021531	1.30×10^{-3}

Error falls as h^2 and then rises: truncation giving way to cancellation. The analytic gradient is correct; the frozen *global* step was too large at those coordinates, and the optimal step is per-coordinate. The plateau check nearly caught it (16.8% spread against a 20% tolerance).

I did not relax the criterion. Releasing those widths requires an amended step rule. Until then the execution path is validated for 10 of 13 widths, which is **incomplete**, and the variance pilot's precondition is unmet.

Appendix C — what the campaign cost, and what it did not authorize

item	GPU device-hours	source
representation matrix (24 paired jobs)	13.27	matrix-summary.json
A3 geometry probe + width gate (5 subs)	0.43	GO_NO_GO-20260918.md §7
tail validation (4 of 4 submissions)	0.77 of 1.5 authorized	validation-final.json
inference benchmark + matrix closeout	<i>not itemized here</i>	—
A1 source characterization	(CPU: 2.0 core-h)	source-multiplicity.json
campaign total	16.0 of 290	RECOMMENDATION-20260918.md

The first three rows are itemized from their own receipts; the total is carried from the campaign recommendation. They do not sum, because the inference benchmark (including one 35-minute timeout that measured nothing) and the matrix closeout are inside the total but are not separately itemized in any receipt I can cite. Stated as a gap rather than closed by arithmetic.

Not launched, and now retired: the *synthetic* four-arm pilot (≤ 30 GPU-h, precondition unmet) and its Stage 2 (≤ 130 GPU-h). **Do not confuse that pilot with stage 4 of the matched comparison** — different sample, different endpoint, different purpose. **Requested instead:** ≤ 70 GPU-h for stages 1-5 of slides 14-16.

Not authorized by anything in this deck: publication adoption, any covariance or systematic, a central-value change, a Gate-6 action, or the construction of C_{ML} . Typed objects and aggregate overflow remain *prospective* representation improvements, exactly as the analysis note already says.

Not done: nothing in this deck has been sent to anyone.